JARVIS Milestone Disposition Report

Track 3.4 — Multi-Session Gateway Integration, Streaming Runtime & Protocol Seam

Date: September 20, 2026 **Host:** Windows 11 Build 26220 (x86_64) **Runtime:** Python 3.13.9 via uv **Status:** Implementation Closed / Frozen

FROZEN MILESTONE DISPOSITION

TRACK 3.4 — CLOSED AS IMPLEMENTATION MILESTONE
VALIDATION PASSED FOR EXECUTED TEST MATRIX (49/49 Dedicated, 666/666 Combined Regression)
FULL QUALIFICATION MATRIX REMAINS OPEN (G01–G27, RH01–RH15)
PRODUCTION INTEGRATION PENDING

1. AUTHORITY MODEL & ARCHITECTURAL BOUNDARIES

Track 3.4 establishes the protocol and streaming integration seam between external clients (Desktop HUD, Web Frontend, Headless CLI, Telegram, Discord, Slack) and the frozen Track 3 foundation. Track 3.4 introduces **zero LLM reasoning, zero autonomous planning, and zero direct execution**.

Subsystem	Canonical Responsibility	Hard Contract Boundary
Track 3.1	Decides / Plans / Authorizes	Sole planning authority; compiles immutable, validated PlanDAGs. Approval is an orthogonal policy gate.
Track 3.3	Schedules / Coordinates / Recovers	12-state node lifecycle, topological execution, fencing tokens, worker leases, crash reconciliation.
Track 3.2	Admits / Sandboxes / Executes / Verifies	Fail-closed execution, sandboxing tiers (Tier 0–3), observation and Level 5 evidence sealing.
EventStore	Authoritative WAL Truth	Append-only SQLite WAL journal; monotonic `rowid` (global_seq) is the sole authoritative reality.
Track 3.4	Protocol / Route / Stream / Delivery	Ingress RPC router, rate limiting, route synthesis, two-tier catch-up, independent transport sequences.

2. RIGOROUS QUALIFICATION MATRIX DISTINCTION

Track 3.4 explicitly differentiates between scenarios **empirically executed and evidenced** versus scenarios remaining **open** for the full qualification milestone:

Verification Suite	Executed & Evidenced Scenarios	Open Qualification Scope
Dedicated Contracts & Seams 38 / 38 PASS	38 tests across 10 modules: Errors, Handshake, Sequence Triad, Auth, Sessions, Idempotency, Dispatcher, Streamer, Catchup, Approvals.	None (Core protocol contracts 100% verified).
Adversarial Suite (G-Series) 7 EXECUTED PASS 20 OPEN	• `G01` : Reconnect during event commit • `G02` : Duplicate chat.send flood idempotency • `G08` : Multi-client approval race resolution • `G14` : Rapid reconnect churn under load • `G17` : Sliding window rate limit enforcement • `G20` : Cross-channel task isolation • `G26` : Sensitive telemetry secret scrubbing	`G03–G07` (Commit races, gaps) `G09–G13` (Cancel races, stale tokens) `G15–G16` (Half-open TCP, socket drops) `G18–G19` (Session focus contention) `G21–G25` (Route drops, retry limits) `G27` (Dead-letter queue overflow)
Real Windows Host (RH-Series) 4 EXECUTED PASS 11 OPEN	• `RH01` : Live Windows TCP loopback socket • `RH06` : Gateway crash/restart zero-redispatch • `RH10` : Subprocess stdout secret scrubbing • `RH15` : Full Telegram ↔ EventStore ↔ Telegram E2E	`RH02–RH05` (TCP stress, half-close) `RH07–RH09` (Process lease arbitration) `RH11–RH14` (Live Web UI & Desktop stress)

3. WORKSPACE REGRESSION VERIFICATION (666 / 666 PASS)

All existing frozen subsystems were executed with zero modifications. Regression arithmetic is verified: **584 (agent_runtime) + 20 (Track 2.2) + 28 (Track 2.3) + 34 (Desktop Host) = 666 PASS**.

Subsystem Suite	Tests Run	Result	Duration	Scope & Notes
Track 3.4 Dedicated Suite	49 / 49	**PASS**	1.98s	38 contracts + 7 adversarial + 4 real host
Track 3.3 Durable Scheduler	74 / 74	**PASS**	6.53s	12-state DAG, fencing tokens, recovery
Track 3.1 Orchestration Foundation	45 / 45	**PASS**	3.82s	Task aggregate, reducers, PlanDAG
Track 3.2 Tool Execution Runtime	25 / 25	**PASS**	3.95s	Tool broker, sandbox tiers, verification
Track 2.2 Persistence & Reboot	20 / 20	**PASS**	9.40s	Real boot detection, WAL durability
Track 2.3 Host Sandbox Harness	28 / 28	**PASS**	14.87s	Restricted tokens, job limits, evidence
Desktop Host Interaction (`tests/`)	34 / 34	**PASS**	242.20s	Live Windows UI, apps, subprocesses
Full `agent_runtime` Suite	584 / 584	**PASS**	12.73s	All agent_runtime modules + Track 3.4
Combined Workspace Verified Total	**666 / 666**	**100% PASS**	**~4.8 min**	**Verified regression total across workspace**

4. LOCKED CONTRACT & INVARIANT DEFINITIONS

- **DeliveryLedger Invariant:** The `DeliveryLedger` guarantees *gateway-side duplicate-dispatch prevention* across process crashes and restarts for identical delivery identifiers. Provider-side exactly-once delivery is not assumed unless the external channel natively supports idempotency or reconciliation.
- **Secret Redaction Invariant:** 100% of tested secret patterns (API keys, bearer tokens, passwords, private keys) are redacted into `[REDACTED_*]` before frame construction; detection across arbitrary unmodeled patterns is not assumed.
- **Sequence Triad Decoupling:** Decoupling of `global_seq` (monotonic SQLite rowid in EventStore), `aggregate_version` (task revision), and `connection_seq` (per-socket counter starting at 1).
- **Server-Authoritative Route Synthesis:** Prevents channel spoofing by constructing `ResponseRoute` strictly from authenticated connection credentials, locking channel identity, conversation, and user ID.
- **Action-Bound Approvals:** SHA-256 canonical digest computed across `(task_id, node_id, revision, capability_id, args_hash, side_effect_class, target_resource)` with atomic SQLite compare-and-set. Duplicate resolutions return cached outcome.
- **Asynchronous Cancellation:** Immediate acknowledgement (`accepted=True, cancellationRequestId`); terminal outcome is derived strictly from Track 3.2 execution reality and streamed via domain events.
- **Two-Tier Catch-Up Handoff:** Atomic snapshot of `MAX(rowid)` before replaying historical range; live events stream after the boundary without duplicates or missed frames.
- **Bounded Backpressure:** Bounded per-connection queues (500 frames); Category B telemetry is dropped under high watermark (80%), while Category A structural events and Category C evidence are preserved. Lagging sockets are disconnected with slow-drain timeout.

5. STATIC ANALYSIS & CI QUALITY GATES

Quality Gate	Verified Target Surface	Result & Compliance
Pyright Typecheck	`packages/agent_runtime/src/jarvis_agent/gateway`	**0 ERRORS** (Strictly typed)
Ruff Linting	All gateway src + 12 dedicated test modules	**ALL CHECKS PASSED**
Ruff Auto-Format	All gateway src + 12 dedicated test modules	**34 FILES COMPLIANT**
Import Boundary Gate	Scanned 984 extension files across repo	**0 VIOLATIONS**
Max Lines Gate	Hard limit 2,000 lines (Target ≤ 700)	**PASS (GATEWAY FILES ≤ 350 LINES)**
Extension Manifest Gate	167 extension manifests	**167 / 167 VALIDATED**
Schema Version Guard	Canonical SQLite schemas	**CANONICAL (0 AUTONOMOUS BUMPS)**

6. MANDATORY GOVERNANCE CONTROLS

LIVE TRAINING	MODEL PROMOTION	MODEL DEPLOYMENT	PROD ACTIVATION
DISABLED	**DISABLED**	**DISABLED**	**DISABLED**

7. IMMEDIATE ACTIONS & NEXT MILESTONE ROADMAP

1. **Preserve Frozen Contracts:** Tracks 3.1, 3.2, 3.3, and 3.4 remain locked; no backports or autonomous changes.
2. **Track 3.4 Full Qualification Campaign (Optional Milestone):** When scheduled, author and execute the remaining 20 adversarial scenarios (G03–G07, G09–G13, G15–G16, G18–G19, G21–G25, G27) and 11 real-host tests (RH02–RH05, RH07–RH09, RH11–RH14).
3. **Gateway Client Adaptation:** Connect the Desktop HUD and modern Web Frontend (`frontend/`) via the validated Track 3.4 WebSocket protocol.
4. **Maintain Governance Freeze:** Production activation controls remain strictly disabled until explicit user-authorized integration milestones.

Signoff: JARVIS Agent Runtime Subsystem Lead **Authority:** Track 3.1 / 3.2 / 3.3 / 3.4 Architecture Committee